

Global Savings Glut:

Inequality, Domestic Asset Markets, and
Economic Growth

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Savings and Inequality are at all time highs in both the world and in the United States. We know that some inequality is good, it is a good incentive for people to move into production of higher utility goods and services and work harder. However, with the recent decades bringing the distribution of wealth to a lopsided 11:1 for the average vs median family in the United States, is there an economic reason why at a certain point inequality is bad? This paper will explore the idea of a global savings glut and in particular, a saving glut of the rich in the United States possibly causing increasingly volatile debt cycles and a long-term economic slowdown. The intuition of this idea follows mainly from the idea of circular flow. Therefore, segmenting where inflation occurs is key to understanding what is happening in the economy.

Imagine the economy as a pile of cash in different areas. Institutions are the pipelines (including fiscal policy), and monetary policy and innovation changes the aggregate. What we are seeing in the economy now is a huge pile of cash savings (including short-term money market and mutual funds) hoarded overseas and in capital markets and not being introduced back into the pipelines. The idea of a global savings glut is not new, economists such as Karl Marx, JA Hobson, and John Maynard Keynes all focused very intently on its consequences on growth and economists such as Ben Bernanke point to the idea as an explanation of the 2008 financial crises and the long-term decline of interest rates. Even though there is no formal or unified definition, I will be defining the Global Savings Glut as: **The phenomena at which net savings surpasses domestic net investment such that the proportion of money that is dead causes inflation in capital markets and lowers overall growth reflected in long term interest rates.** “Dead” is defined as “cash or allocated capital towards consumption from debt.” But why does this happen and what are the effects? We will look at empirical evidence linking

inequality to giant sums of cash hoarding, how this effects human behavior, and what it means long term for growth and short and long run welfare in society. Specifically, we will look at the global impacts on the United States domestic economy, where all the factors I describe will converge. Two concepts are key here; 1) understanding circular flows across the economy and how they connect and 2) the first assumption relies on institutions allocating resources to future production of goods and services that provide a sufficient level of utility (essentially enough demand for the product not to waste increased production).

Savings vs Investment Preferences

Both the National Income and Product Accounts and household wealth reported in the Financial Accounts of the Federal Reserve point to an increase in the savings rate by the 1% of 3% between the 1970's and post-2000¹. This is substantial as savings surpassed net domestic investment in 2000 and most of these savings show themselves in financial markets². As economic students we tend to believe that financial markets allocate resources efficiently into investments, however, the evidence as shown in **The Savings Glut of the Rich** by Straub, Sufi, Mian³, identifies that after removing intermediaries and looking at the US Financial Accounts, almost 2/3 of rise in asset accumulation by the top 1% has gone into claims against government and household debt. By using a panel of states to control for financial de-regulation, wages paid by employers, tax incentives, and pre-trends, the evidence shows that increases in top income

¹ Appendix [1]

² Appendix [2]

³ Mian, A., Straub, L., & Sufi, A. (2020). The Saving Glut of the Rich and the Rise in Household Debt. doi:10.3386/w26941

earning are consistent with an increased desire to save. This relationship between macroeconomics and the financial sector is puzzling. If investment is decreasing and governments and the bottom 90% are dissaving, then is the non-production of the financial sector leading to any long-term detriments?

An important question to ask however is why did the bottom 90% of income earners decide to dissave? Cynamon and Fazzari⁴ show that rising inequality starting in the 1980's forced the bottom 90% to either cut consumption or borrow further to negate short-term payments owed on their balance sheets. With the subsequent boom in household debt levels until the Great Recession, we see that they chose the latter. Since then, real household demand has grown at the same pace as the pre-recession trend, however, at a difference in intercept of about 1.5 trillion 2009 dollars and noted that into 2015 it still held⁵. If we assume that recessions are truly temporary, this persistent deficit below potential gives cracks to the idea that long-term economic growth is mostly determined by long-run aggregate supply. With household debt increasing even more substantially with growing inequality, we must conclude that the availability of debt financing is convincing by itself for consumers to keep their optimal level of consumption. We see that inequality is forcing either austerity or increasingly volatile cycles of debt. Household debt is again at an all-time peak (pre-COVID) and with still stagnant wages there's no real increase in purchasing power except that the drop in interest rates have allowed the eventual re-payment schedule of debt to extent in the future. In other words, too

⁴ Cynamon, Barry Z., and Steven M. Fazzari. "Rising Inequality, Demand, and Growth in the US Economy." SSRN Electronic Journal, 2015, doi:10.2139/ssrn.2570506.

⁵ Appendix [3]

few people having too much savings is causing a demand drag on the entire middle and lower class by hindering the demand generation process. The historical trend of inequality could be causing a disruption to the economy via circular flow because higher incomes have a lower propensity to consume, however, Cynamon and Fazzari note that the timing tells a different story. As income inequality started to rise in the 1980's, household demand to cash ratio did not decline, however, the simple interpretation at face value does not include financial dynamics. With the massive amount of borrowing available as discussed in Straub, Amir, and Mian's work, the savings glut can postpone unsustainable trends. Cynamon and Fazzari use a panel to assess the effect of income distributions on the Keynesian Multiplier and find that the US could be around 10% or more below potential if the income distribution followed the trends in the 1960's-1970's⁶.

The increase in inequality is not only attributed to domestic United States economic activity. The increase in savings by the 1% in, particularly, oil nations and emerging Asian markets, has caused massive amounts of capital flows into the United States in order to finance the capital account and external deficit. Using the FRED database, we see that net capital flows is 0 until the 80's where it starts increasing linearly from 0 to about a max of 111 billion dollars in 1995 and from there it reaches a peak of 726 billion dollars right before the financial crash and has since been increasing again with a weighted average of about 200 billion dollars since the trough⁷. The capital accounts of the US and China were almost mirrored until the early 2010's where demand from other countries started soaring. Indeed, Bertaut, DeMarco, Kamin,

⁶ Appendix [4]

⁷ <https://fred.stlouisfed.org/series/BOPI>

and Tyron⁸, show that selling of portfolio debt has made up the bulk of the financing of the US current account deficit and that Agency backed (Government guaranteed) and corporate bonds have been sold to large institutional investors, again confirming that the global 1% are using their savings to finance household and government borrowing in the United States. As they further examine in their paper, the rise of surplus funds into the United States were exogenous, independent of financial developments, thus re-enforcing the case that these surplus funds were responsible for decreasing interest rates. In fact, they find using an OLS regression that builds off Warnock and Warnock (2009,) using outstanding US Treasuries and Agencies data from June 2007, that a \$100 billion dollar increase in the foreign official inflows decreases treasury yields by 11 basis points. Using a panel of similar studies, we can confirm their findings are in a consistent range⁹.

As discussed, prior, before the Great Recession, foreign capital inflows were going almost entirely into agency backed securities, mostly mortgage backed securities (MBS) and US Treasury Bills¹⁰. As mentioned, the increase in financing by the US 1% and the massive increase in savings globally drove down interest rates, however, the saving countries (Emerging Asia and Oil nations) that poured their savings into the United States did not expect to end up in risky investment at all really. Bernanke et al, 2011, explains how because all these funds were directed into fixed income securities, the yield on safe assets decreased so much that private investors started demanding alternative investments to match historical expectations. In fact,

⁸ Bertaut, C., Demarco, L. P., Kamin, S., & Tryon, R. (2011). ABS Inflows to the United States and the Global Financial Crisis. doi:10.3386/w17350

⁹ Appendix [5]

¹⁰ Appendix [6]

the reason why European institutions ended up with much of the private-labeled asset backed securities (ABS) is because the emerging Asian countries and oil exporters took most of the Agency-backed securities off the market for which the increase in supply could only marginally make up for. On the eve of the first financial crash during the Great Recession, the share of foreign holdings of Agency backed securities were at 31% and ABSs were at 24%. Out of Corporate bonds held by foreigners, nearly half were securitized into ABSs and out of the non-ABS holdings, a majority were in alternative assets such as structured cash flows, floating rate notes, and financial debt obligations¹¹. The appetite for risk manifested itself through these financial intermediaries being compensated for high yields but not penalized for tail risk when the eventual crash happened.¹² Essentially the boom in foreign funds into the US had to go into a specific sector and because of easy financing through the 1% of the US mixed with the need of assets abroad, along with the governments deregulation of the financial sector, subprime lending created the perfect mix of an American debt-trap fueling the global asset market.

Net Household and Government Debt Positions

Not only have the bottom 90% started borrowing more since the 1980's, but they have also reduced asset accumulation that is a direct claim on other's borrowing. In Straub's research he measures household debt position between the wealth groups over time and comes to the conclusion that at the peak before the Great Recession the top 1% were owed a net 12% of GNP, while the bottom 90% owed 39% of GNP¹³ and they also include multiple

¹¹ Albertus, Bertaut, and Curcuru "Has the Crisis Changed Foreign Positions in U.S. Securities?" Federal Reserve staff working paper 2010.

¹² Kashap, Rajan, and Stein, 2008; Arteta et al., 2010

¹³ Appendix [7]

approaches to confirm robustness. In their research, Straub, Sufi, and Mian identify that 30% GNP owed from the bottom 90% were financed by the top 1%. Through financial unveiling, their finding for the 3% average increase in government and household debt by year, is that 1.7% is owned by the top 1% in the United States and 2% is owned by the rest of the world (3.7%-debt owed by bottom 90% to the government equals 3%)¹⁴. Now we conclude that there is no question that an increase in savings by the American 1% has been mirrored by dissaving's by the bottom 90% and the government. Straub, Sufi, and Mian's state level analysis allows us to confirm that the increase in top income shares has been accompanied by an increased propensity to save. As shown by this research, an increase in income inequality has completely transformed the level of wealth inequality which links low interest rates to indebtedness as Cynamon and Fazzari explained in their paper.

Cash Hoarding, Returns to Capital, and the Billionaires Jet:

Recessions are caused by a mix of debt and bad investments. Like we saw in the Great Recession, hoarding of Agency Debt caused yields to drop and prices to skyrocket but the underlying fact is that the money had no where better to go. This is what is called a **Misguided Market**, where essentially the demand for returns diverges the price of an asset from its utility provided. In these scenarios, genuinely good investments dry up, and when banks only provide a small interest rate, demand for securities shoots through the roof. While we've seen the cash ratio for the SP500 double since the early 2000's from around 9% to 18%, the Capex

¹⁴ Appendix [8]

(percentage of capital expenditures as a percent of GDP) has declined by 3%¹⁵ since the 1970's, mirroring almost exactly the increase in savings by the American 1%. Two great quasi-natural experiments show the effect of introducing more cash into the economy exogenously and the effects on domestic investment and GDP growth. The 2004 American Jobs Creation Act allowed a major tax cut on foreign repatriated cash. Clausing (2005) notes that firms were able to take advantage of the tax cut by only pursuing projects they had already intended on and then shifted almost all other cash to dividends and share buybacks¹⁶ and Dharmapala estimated that less than one cent on every dollar was reinvested¹⁷. Many would think that at least in the future more funds could be deployed domestically, Dharmapala addresses directly how the expectation of future repatriation deals from the government caused more hoarding overseas. In other words, a classic solution posed to the problem of cash hoarding fails because executives expect to be alive to reap the benefits of the next repatriation deal.

In the last ten years we've seen a total of 4.18 trillion dollars of share buybacks and this has manifested itself in a 65% increase in the sp500 since 2017, even amid the coronavirus, the stock market continues to soar with the flush of money put in. The problem of course is that a staggering amount of the market cap in the stock market does not even plan to turn a profit, like Amazon, Uber, and many technology companies. In fact, according to initial public offering

¹⁵ Brennan, P. (2020, July 01). US companies hoarding cash seen dragging on recovery until coronavirus beaten. Retrieved November 17, 2020, from <https://www.spglobal.com/marketintelligence/en/news-insights/latest-news-headlines/us-companies-hoarding-cash-seen-dragging-on-recovery-until-coronavirus-beaten-59263746>

¹⁶ Clausing, K. (2005). Tax Holidays (and Other Escapes) in the American Jobs Creation Act. *National Tax Journal*, 58(3), 331-346. Retrieved from <http://www.jstor.org/stable/41790272>.

¹⁷ Dharmapala, Dhammika & C. Fritz Foley & Kristin J. Forbes, 2011. Watch What I Do, Not What I Say: The Unintended Consequences of the Homeland Investment Act. *Journal of Finance*, American Finance Association, vol. 66(3), pages 753-787, June.

(IPO) data compiled by Jay Ritter, in 2019 78% of IPO listings did not turn a profit, and this is trending upwards, compared to a decade average of 32%¹⁸. If the large piles of cash cannot even fund companies at the tune of billions of dollars per year to profitability in the near future, then it is a reasonable conclusion that we could be hitting diseconomies of scale. If there are less and less profitable investment opportunities, when the next Uber or Airbnb comes onto the market, the ultra-wealthy investment institutions and billionaires are going to start a wild bidding war in the hope that it will be the next Amazon. Now the reader of this paper might have a genius widget invention or a new cupcake design that can turn a 500% profit in the first year with an investment of \$10,000 but that measly \$50,000 profit won't even cover the jet fuel it took to get to the pitch and back. Even with the much higher returns to capital of small-scale projects. We can see how this might have an adverse effect on small business as they get crowded out. The compounding effect of wealth plays out here to the advantage of large-scale business as according to the US's official "small business GDP" report¹⁹ large businesses reported 2.5% of annualized growth vs 1.4% for small business, in addition the percent of small business relative to GDP fell from 48% to 43.5% in 2014. This correlation can be understood by revisiting the paper by Cynamon and Fazzari, who demonstrate the long-term demand strain on the economy caused by debt cycles. As small businesses do not have the cushion that large businesses have, the volatile debt cycles in the economy cause a disruption over the long term.

¹⁸ <https://site.warrington.ufl.edu/ritter/files/IPOs2019Statistics.pdf>

¹⁹ Kobe, K. (2018, December). Small Business GDP 1998–2014. Retrieved from <https://cdn.advocacy.sba.gov/wp-content/uploads/2018/12/21060437/Small-Business-GDP-1998-2014.pdf>

Conclusion

The Global Savings Glut phenomena is plaguing the United States with new problems for the developed nation. With savings on the rise domestically, we see the bottom 90% of American's forced to either borrow beyond their means or cut consumption dramatically. The effect on the economy, is a lower marginal propensity to consume over the long run and a demand drag. The economy in the United States is trending in a favorable direction towards the wealthy and high income and what we've seen over the past three-four decades is a 3% increase in savings preferences and a large stockpile of overseas money and $\frac{2}{3}$ of that money is being used to finance household and government debt. It should be noted that the 17% or so savings rate for the US is misleading as Straub, Amir, and Mian show. The bottom 90% are deeply in debt and the wealth accumulation of the top is larger than we have ever seen. Global capital inflows pose new questions on how to cope with lower returns, as with the Great Recession, incentives can perverse easily and Dharmapala gives light to how difficult it really is to get cash that leaves the economy back. New questions need to be posed to the financial sector. It would be important to assess if they provide the best outcomes for the economy with the major decline of net domestic investment. Prescriptively, higher taxes are necessary to make public investments as we have seen the equity markets fail to turn a profit in a surprising wide range of market capitalization. A major obstacle is that overpricing must be realized to re-proportion income distribution or inflation will continue to re-affirm fixed assets and equities.

Journals:

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- 2: Bertaut, C., Demarco, L. P., Kamin, S., & Tryon, R. (2011). ABS Inflows to the United States and the Global Financial Crisis. doi:10.3386/w17350
- 3: Cynamon, Barry Z., and Steven M. Fazzari. "Rising Inequality, Demand, and Growth in the US Economy." SSRN Electronic Journal, 2015, doi:10.2139/ssrn.2570506.
- 4: Dharmapala, Dhammika & C. Fritz Foley & Kristin J. Forbes, 2011. Watch What I Do, Not What I Say: The Unintended Consequences of the Homeland Investment Act. Journal of Finance, American Finance Association, vol. 66(3), pages 753-787, June.

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- 1: <https://fred.stlouisfed.org/series/BOPI>
- 2: Kashap, Rajan, and Stein, 2008; Arteta et al., 2010
- 3: Bernanke, B., Bertaut, C., Demarco, L. P., Kamin, S., International Capital Flows and the Returns to Safe Assets in the United States, 2003-2007
- 3: Albertus, Bertaut, and Curcuru "Has the Crisis Changed Foreign Positions in U.S. Securities?" Federal Reserve staff working paper 2010.
4. Brennan, P. (2020, July 01). US companies hoarding cash seen dragging on recovery until coronavirus beaten. Retrieved November 17, 2020,
- 5: Clausing, K. (2005). Tax Holidays (and Other Escapes) in the American Jobs Creation Act. National Tax Journal, 58(3), 331-346. Retrieved from <http://www.jstor.org/stable/41790272>
- 6: <https://site.warrington.ufl.edu/ritter/files/IPOs2019Statistics.pdf>

7: <https://site.warrington.ufl.edu/ritter/files/IPOs2019Statistics.pdf>

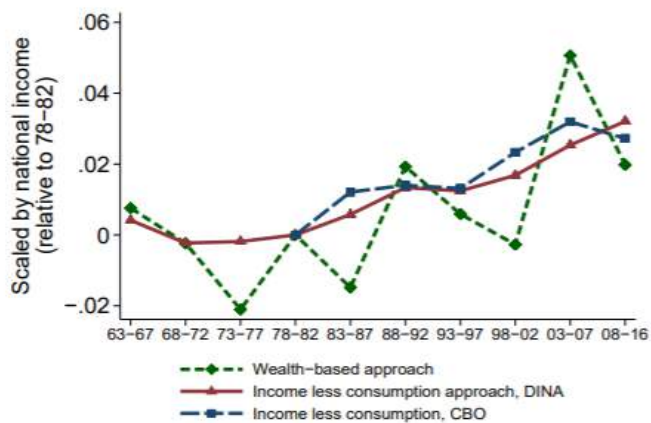
8: Kobe, K. (2018, December). Small Business GDP 1998–2014. Retrieved from

<https://cdn.advocacy.sba.gov/wp-content/uploads/2018/12/21060437/Small-Business-GDP-1998-2014.pdf>

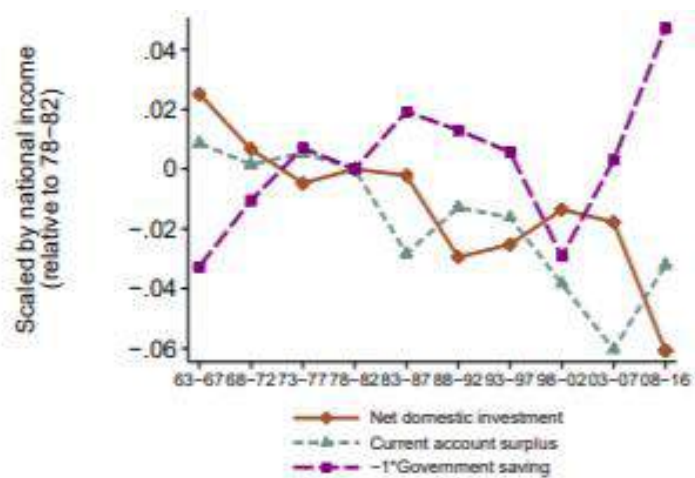
Appendix:

[1]

Figure 2: Savings by the Top 1%

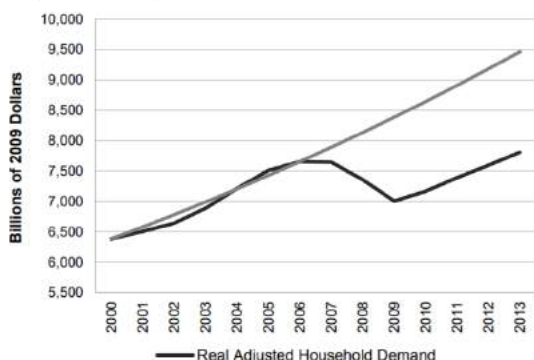


[2]



[3]

Figure 1 – Real Adjusted Household Demand and Pre-Recession Trend



[4]

Table 1 – Calibrated Effects of Change in Distribution on the Multiplier

Source for MPC Parameters	Bottom 95% MPC, α_L	Top 5% MPC, α_H	Impact of Change in Shares ($\Delta\lambda_H$) on the Multiplier, ΔM
Johnson <i>et al.</i> (2006)	0.40	0.20	-3.7%
CF	0.92	0.82	-9.5%
Bakker and Felman (2014)	0.95	0.65	-13.6%

Note: These results assume marginal tax rates for the bottom 95% (τ_L) and top 5% (τ_H) of 0.20 and 0.40 respectively. The change in the pre-tax income share of the top 5% ($\Delta\lambda_H$) is 22.8% to 36.6%.

[5]

$$R_{10,t} = f\left(R_{10,t-1}, \bar{Y}_t^e, \pi_{t+10}^e, rp_t, R_{ff,t}, \frac{F_t}{D_t}\right) \quad (1)$$

$R_{10,t}$: 10-year U.S. Treasury yield, nominal

$\bar{Y}_t^e$: expected GDP growth (blue chip)

π_{t+10}^e : expected inflation rate (10-year ahead Michigan survey)

rp_t : risk premium (36-month standard deviation of changes in Treasury yield)

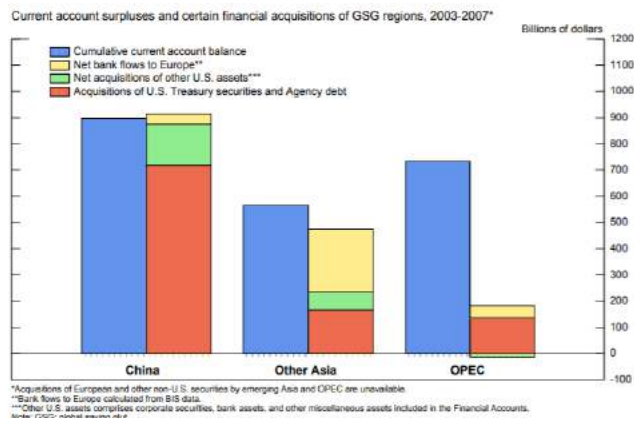
$R_{ff,t}$: federal funds rate

F_t : foreign official holdings of U.S. Treasuries and Agencies

D_t : total outstanding marketable U.S. Treasuries and Agencies

Study	Impact of \$100 billion in U.S. Treasury/Agency purchases on Treasury yields (basis points)
Bertaut et al. (2011): Reduced form regression	-11
Bertaut et al. (2011): Portfolio balance model (section V, below)	-15
Warnock and Warnock (2009)	-15 to -32
Beltran et al. (2011)	-49
Rudebusch, Swanson and Wu (2006)	no significant effect
Gagnon et al.(2010)	-6
D'Amico and King (2011)	-10
Hamilton and Wu (2010)	-4

[6]



[7]

Figure 4: Saving across the Distribution

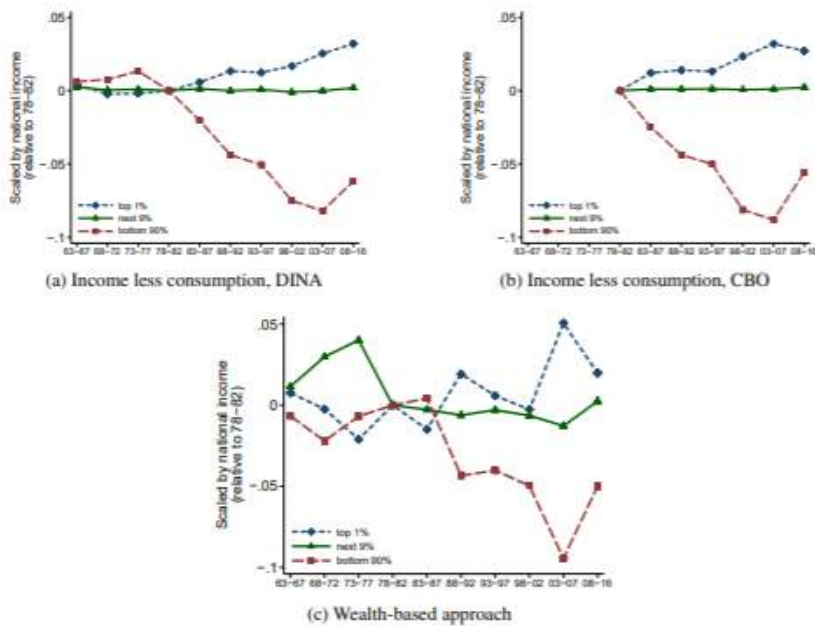
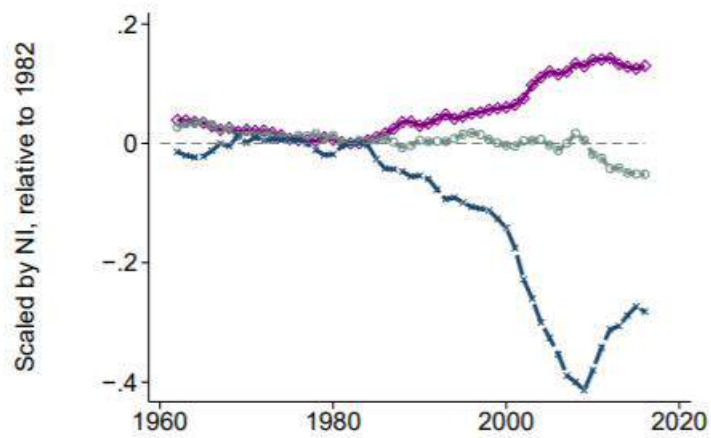


Figure 9: Net Household Debt across Wealth Distribution Relative to 1982



[8]

Figure 10: Sources of Financing for Rise in Government and Household Debt

